

INNOVATION PROPOSAL

Enhancing Kenya Power's Prepaid Electricity Experience Through Bluetooth-Enabled Smart Customer Interface Unit (CIU) Integration

Prepared by

Kahindi Padii Gandi

Android & Embedded Systems Developer

KCA University

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Executive Summary

Kenya Power has made significant progress in digital transformation through initiatives such as the MyPower mobile application and the Tuende Digital campaign. Customers can conveniently purchase prepaid electricity tokens through multiple digital payment channels, reducing the need to visit physical payment locations.

Despite these advancements, customers continue to experience challenges after purchasing electricity tokens. The current prepaid system still requires manual entry of lengthy token numbers into the Customer Interface Unit (CIU). In addition, customers must physically access the CIU to monitor remaining electricity units, daily consumption, and other meter information. This often results in customers unexpectedly running out of electricity due to lack of timely monitoring or forgetting to purchase additional tokens.

This proposal presents a Bluetooth-enabled enhancement to the existing prepaid electricity infrastructure. The proposed solution integrates a Bluetooth communication module with the existing Customer Interface Unit, allowing secure communication with the official Kenya Power MyPower mobile application. The solution enables customers to monitor electricity information from their smartphones and automatically transfer purchased prepaid tokens to the CIU while preserving the existing prepaid infrastructure.

The proposal does not seek to replace Kenya Power's current prepaid metering system. Instead, it introduces a backward-compatible enhancement that improves customer convenience, increases digital engagement, and supports Kenya Power's ongoing digital transformation strategy.

1. Background

Kenya Power has continuously invested in improving customer experience through digital innovation. The introduction of prepaid electricity meters and the MyPower mobile application has significantly simplified electricity purchases for prepaid customers.

Although customers can now purchase electricity tokens from virtually anywhere using mobile payment platforms, the final stage of the process still requires manual interaction with the Customer Interface Unit. Customers must physically enter lengthy prepaid token numbers before electricity credit can be added to their meters.

Additionally, while modern Customer Interface Units provide valuable information such as remaining units, voltage, and electricity consumption, this information remains available only on the CIU display. Customers are therefore unable to monitor their electricity usage remotely or receive proactive notifications regarding low electricity balances.

As Kenya Power continues to expand its digital services, integrating mobile technology with the existing prepaid infrastructure presents an opportunity to further improve customer convenience without requiring replacement of existing metering infrastructure.

2. Problem Statement

The existing prepaid electricity system requires customers to manually monitor their electricity usage and manually enter prepaid token numbers into the Customer Interface Unit after every purchase.

This process presents several challenges, including:

- Customers forgetting to monitor their remaining electricity units.
- Unexpected electricity interruptions caused by depleted prepaid balances.
- Inconvenience associated with manually entering lengthy prepaid token numbers.
- Lack of real-time access to electricity information through mobile devices.
- Limited integration between the existing MyPower mobile application and Customer Interface Units.

Although the current infrastructure reliably supports prepaid electricity services, opportunities exist to further improve customer convenience through enhanced mobile integration.

3. Research Objectives

The primary objective of this proposal is to demonstrate how Bluetooth Low Energy technology can improve customer interaction with Kenya Power's existing prepaid electricity infrastructure.

Specific objectives include:

- Reduce customer inconvenience during prepaid token loading.
 - Improve customer awareness of electricity consumption.
 - Enable secure communication between Customer Interface Units and the MyPower mobile application.
 - Increase customer engagement with Kenya Power's digital services.
 - Demonstrate a backward-compatible enhancement requiring minimal infrastructure changes.
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4. Existing System

The current prepaid electricity workflow consists of the following steps:

1. Customer purchases prepaid electricity tokens.
2. Kenya Power generates a prepaid token.
3. Customer receives the token via SMS or the MyPower application.
4. Customer manually enters the prepaid token into the Customer Interface Unit.
5. The Customer Interface Unit forwards the token to the electricity meter.
6. The meter validates the token and updates the customer's electricity balance.

While this process is reliable, the final manual step introduces unnecessary inconvenience and potential user errors.

5. Proposed Solution

This proposal introduces a Bluetooth-enabled communication module integrated with the existing Customer Interface Unit.

The module establishes secure communication between the Customer Interface Unit and the official Kenya Power MyPower mobile application.

The proposed solution enables customers to:

- View remaining electricity units from their smartphones.
- Receive low-balance notifications.
- Monitor electricity consumption.
- View selected electrical parameters available from the Customer Interface Unit.
- Automatically transfer purchased prepaid tokens from the MyPower application to the Customer Interface Unit.

Manual token entry remains available at all times, ensuring full compatibility with the existing prepaid electricity process.

6. System Architecture

The proposed solution introduces an additional communication layer between the Customer Interface Unit and the MyPower mobile application without modifying Kenya Power's backend systems or prepaid token generation process.

The architecture consists of the following components:

- Existing Kenya Power prepaid infrastructure.
- Customer Interface Unit (CIU).
- Bluetooth Communication Module.
- MyPower Mobile Application.

The Bluetooth Communication Module acts as a secure intermediary between the mobile application and the Customer Interface Unit, facilitating two-way communication while maintaining compatibility with the existing infrastructure.

7. Prototype Overview

A functional proof-of-concept prototype has been developed to demonstrate the feasibility of the proposed solution.

The prototype demonstrates:

- Secure Bluetooth communication.
- Mobile application connectivity.
- Real-time information exchange.
- Automatic token transfer simulation.
- Customer Interface Unit interaction.

The prototype validates the core concept and provides a foundation for future integration with production Customer Interface Units.

8. Benefits

Benefits to Customers

The proposed solution provides several customer-focused improvements, including:

- Improved convenience.
- Reduced likelihood of unexpected electricity interruptions.
- Real-time access to electricity information.
- Reduced manual data entry.
- Enhanced accessibility for elderly and physically challenged users.
- Improved overall user experience.

Benefits to Kenya Power

The proposed solution supports Kenya Power's strategic objectives by:

- Supporting the Tuende Digital initiative.
 - Increasing adoption of the MyPower mobile application.
 - Improving customer satisfaction.
 - Introducing additional digital services without replacing existing prepaid infrastructure.
 - Providing a platform for future smart electricity services.
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9. Security Considerations

Security remains a critical component of the proposed solution.

The proposal recommends:

- Secure Bluetooth pairing.
- Device authentication.
- Encrypted communication.
- Customer authorization before automatic token transfer.
- Continued support for manual token entry.

The proposed system is designed to enhance customer convenience while preserving the security principles of the existing prepaid electricity infrastructure.

10. Implementation Strategy

A phased implementation strategy is recommended.

Phase 1

Proof of Concept (prototype)

Development and validation of Bluetooth communication and mobile integration.

Phase 2

Pilot Program

Controlled testing using selected Customer Interface Units and volunteer customers.

Phase 3

Technical Integration

Collaboration with Kenya Power engineers and Customer Interface Unit manufacturers to integrate the communication module with production hardware.

Phase 4

Commercial Deployment

Gradual rollout as an optional enhancement for prepaid customers.

11. Challenges and Considerations

Several factors should be considered during implementation.

These include:

- Compatibility across different Customer Interface Unit manufacturers.
- Security certification.
- Bluetooth communication reliability.
- Hardware integration.
- Regulatory compliance.

These challenges can be addressed through collaboration with Kenya Power and Customer Interface Unit manufacturers during the implementation phase.

12. Future Enhancements

Future versions of the proposed solution may include:

- Predictive electricity consumption analysis.
- Personalized electricity usage reports.
- Smart home integration.
- Automated low-balance reminders.
- Remote diagnostics.
- Enhanced customer analytics.
- AI
- WiFi integration for remote access.

13. Conclusion

The proposed Bluetooth-enabled Customer Interface Unit represents a practical enhancement to Kenya Power's existing prepaid electricity ecosystem.

Rather than replacing current infrastructure, the proposal introduces a backward-compatible communication layer that improves customer convenience, strengthens engagement with the MyPower mobile application, and supports Kenya Power's digital transformation objectives.

A proof-of-concept prototype has demonstrated the technical feasibility of the proposed approach. The next stage would involve collaboration with Kenya Power and Customer Interface Unit manufacturers to evaluate integration into production systems while maintaining the reliability and security of the existing prepaid electricity infrastructure.

The proposed solution aims to contribute to a more connected, convenient, and customer-centric prepaid electricity experience for Kenya Power customers.

Note:

The project cost is to be determine for I am not sure of the full price.